

MCA Question Presenter User Guide

Instructor guide for live MCA practice sessions

Purpose

MCA Question Presenter is a local desktop tool for running live MCA practice sessions. It lets an instructor screen-share questions, move through a shuffled deck, and reveal the correct answer only when ready.

Simple by design: no login, no database, no web server, no student grading, and no WordPress dependency.

Included Banks

Bank	File	Use
Navigation	<code>samples/oow-navigation-bank-150.csv</code>	Navigation, chartwork, lights, signals, rules, and related OOW practice.
Stability	<code>samples/oow-stability-bank-150.csv</code>	Stability, loading, buoyancy, free surface, and related OOW practice.

The Navigation bank loads automatically when the app starts. Use **Load Selected** to switch to the Stability bank.

Install On Linux

```
sudo apt update
sudo apt install -y git python3 python3-tk

mkdir -p ~/code
cd ~/code
git clone https://github.com/Dash1971/mca-question-presenter.git
cd mca-question-presenter

python3 -m unittest discover -s tests
python3 -m compileall presenter tests
python3 presenter/mca_question_presenter.py
```

Start A Teaching Session

1. Launch the app with `python3 presenter/mca_question_presenter.py`.
2. Use the private **Instructor controls** window to drive the session.
3. Share only the **MCA Question Display** window in Zoom, Teams, or Google Meet.

4. Press **Next** to show the first question.
5. Discuss the question, then press **Reveal / Hide** when ready.
6. Press **Next** to continue.

Instructor Controls

Control	What It Does
Load CSV...	Open a custom CSV question bank from your computer.
Show Display	Bring the presentation window back to the front.
Topic	Filter the current bank by topic_tag.
Previous	Return to the previous question.
Next	Move to the next question.
Reveal / Hide	Show or hide the correct answer.
Reshuffle	Randomize the current topic or full bank again.
Load Selected	Load one of the bundled banks.

Keyboard Shortcuts

Right arrow Next question	Left arrow Previous question
Space or Enter Reveal or hide answer	F11 / Escape Toggle or leave fullscreen

Recommended Screen-Share Setup

Keep **Instructor controls** private and share **MCA Question Display** with students. If the display window disappears behind other windows, click **Show Display**. For a cleaner lesson view, press **F11** while the display window is active.

Question Bank Format

Required columns:

```
question_text,option_a,option_b,option_c,option_d,correct_option
```

Recommended full format:

portal,question_text,option_a,option_b,option_c,option_d,correct_option,topic_tag,is_active,sort_order

- correct_option must be A, B, C, or D.
- topic_tag controls the topic filter.
- sort_order controls the stable load order before shuffling.
- is_active values of 0, false, no, inactive, or blank are skipped.

Troubleshooting

Problem	Fix
ModuleNotFoundError: tkinter or no window opens	Install Tkinter with <code>sudo apt install python3-tk</code> .
The selected bank has no questions	Check that is_active is not blank, 0, false, no, or inactive.
A CSV will not load	Confirm all required columns exist and every row has A-D options.
The wrong answer is highlighted	Check the row's correct_option value.
Students can see the controls	Share only the MCA Question Display window.

Scope

This app is for live teaching and practice. It does not record student answers, calculate scores, manage course access, or run online exams.